06/06/20

46

# Logic

**Proposition**: A declarative sentence to which we can assign one and only one of the truth values.

two types of propositions:

(i) Atomic propositions: Can't be divided into 2 or more propositions

(ii) Compound propositions: 2 or more propositions connected by a connective

## Connectives:

(i) Conjunction ($\wedge$) (and / but):

* True when all atomic propositions are true
* Commutative & associative

Eg: $p \wedge q$

(ii) Disjunction ($\vee$) (Inclusive OR):

* True when atleast one atomic proposition is true
* Commutative & associative
* $p \vee q$

(iii) Single Implication ($\rightarrow$)

* It is written as $p \rightarrow q$
* It can be read as any of the below ways
  - → if p, then q
  - → if p, q
  - → q, if p
  - → q, when p
  - → q, whenever p

Truth table:

| p | q | $p \rightarrow q$ |
|---|---|---|
| F | F | T |
| F | T | T |
| T | F | F |
| T | T | T |

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* → q unless ~p
  → p implies q
* → p only if q
* → p is sufficient condition for q
* → q is necessary condition for p
* → q follows from p
* → q provided p
* → q is consequent of p

* In $p \to q$

  p is called antecedent (or) premise (or) hypothesis

  q is called consequent. (or) conclusion

* Implication is right associative $\boxed{p \to q \to r \cong p \to (q \to r)}$

(iv) Bicondition (↔)

* $p \leftrightarrow q$

  read as
  - → if and only if
  - → p iff q
  - → p exactly when q
  - → if p then q and conversly
  - → p is neccessary and sufficient for q.

Truth table:

| p | q | $p \leftrightarrow q$ |
|---|---|---|
| F | F | T |
| F | T | F |
| T | F | F |
| T | T | T |

* $\sim(p \leftrightarrow q) \cong p \oplus q$ (i.e., XOR operation)

## Modifier:

**Negation: (~)**

It is a modifiers whose application inverts the truth value.

**Note:**

* $(p \leftrightarrow q) \cong (p \to q) \wedge (q \to p)$
* $p \to q \cong \sim p \vee q \cong \sim q \to \sim p$

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**Note:**

The converse of $p \to q$ is $q \to p$
inverse of $p \to q$ is $\sim p \to \sim q$ } equivalent

contrapositive of $p \to q$ is $\sim q \to \sim p$

$$p \to q \equiv \sim q \to \sim p$$

**Tautology (valid)**: A compound proposition that is always true.

**Contradiction**: A compound proposition that is always false.

**Contingency**: Neither Tautology nor Contradiction

**Satisfiability**: Not a contradiction (i.e., Tautology or Contingency)

**Taulogical Implication:**

Let P, Q be compound propositions,

The implication $P \to Q$ is said to be tautological implication iff $P \to Q$ is a tautology.

It is written as $P \Rightarrow Q$

**Logical equivalence:**

Let P, Q be compound propositions

if P & Q have same truth tables we say P is equivalent to Q. It is denoted as $(P \Leftrightarrow Q)$ or $(P \equiv Q)$

In other words

$(P \Leftrightarrow Q)$ iff $(P \leftrightarrow Q)$ is a tautology

**Note:**

* Order of precedences of connectives is

$$(\sim) > (\wedge) > (\vee) > (\to) > (\leftrightarrow)$$

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# Equivalences:

* $p \wedge T \equiv p$, $p \vee F \equiv p$ } Indentity laws

* $p \vee T \equiv T$, $p \wedge F \equiv F$ } Dominating laws

* $p \vee p \equiv p$, $p \wedge p \equiv p$ } Idempotent laws

* $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$, $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ } Distributive laws

* $\sim(p \wedge q) \equiv \sim p \vee \sim q$, $\sim(p \vee q) \equiv \sim p \wedge \sim q$ } Demorgan's laws

* $p \vee (p \wedge q) \equiv p$, $p \wedge (p \vee q) \equiv p$ } Absorption laws

* $p \vee \sim p \equiv T$, $p \wedge \sim p \equiv F$ } Negation laws

* $p \rightarrow q \equiv \sim p \vee q \equiv \sim q \rightarrow p$

* $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow r) \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$

* $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

  $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$

  $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$

  $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Tip:
In left side ~~p~~ formulas
if $p$ is common in both then symbol is same
if $r$ is common, then symbol changes

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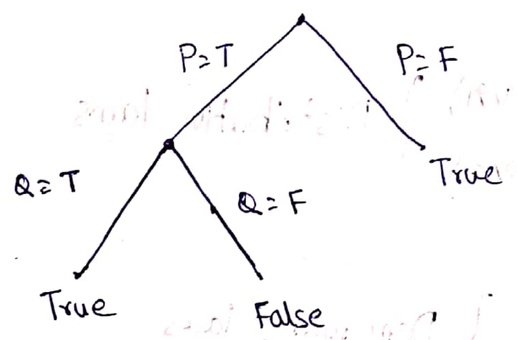
## Quines's Method :

* This method is used to find whether the given compound proposition is valid or contradiction or ~~st~~ satisfiable or contingency.

* Here we use appropriate equivalences and reduce compound proposition by substituting truth values

Eg:

$$P \rightarrow (Q \wedge P)$$




$\therefore$ contingency

## Argument (Inference):

If a set of premises $\{P_1, P_2, P_3 \cdots P_n\}$ yield another, proposition Q (conclusion) then whole process is called argument or, inference.

It is denoted as

$\{P_1, P_2 \cdots P_n\} \vdash Q$ (or), $\dfrac{P_1, P_2 \cdots P_n}{Q}$ (or), $\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \\ \hline \therefore Q \end{array}$

(or)

$\{P_1, P_2 \cdots P_n\} \Rightarrow Q$

(or)

$(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \rightarrow Q$

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## Rules of Inference (Tautological implications)

① $$\begin{array}{l} p \to q \\ p \\ \hline \therefore q \end{array}$$ Modus ponens

② $$\begin{array}{l} p \to q \\ \sim q \\ \hline \sim p \end{array}$$ Modus tollens

③ $$\begin{array}{l} p \vee q \\ \sim p \\ \hline q \end{array}$$ Disjunctive syllogism

④ $$\begin{array}{l} p \to q \\ q \to r \\ \hline p \to r \end{array}$$ Hypothetical syllogism

⑤ $$\begin{array}{l} p \\ \hline p \vee q \end{array}$$ Addition

⑥ $$\begin{array}{l} p \wedge q \\ \hline \therefore p \\ \therefore q \end{array}$$ Simplification

⑦ $$\begin{array}{l} p \\ q \\ \hline p \wedge q \end{array}$$ Conjunction

⑧ $$\begin{array}{l} p \vee q \\ \sim q \vee r \\ \hline p \vee r \end{array}$$ Resolution

## Fallacies (Invalid arguments):

(i) Fallyacy of assuming converse:

$$\begin{array}{l} p \to q \\ q \\ \hline \therefore p \end{array}$$

(ii) Fallacy of assuming inverse:

$$\begin{array}{l} p \to q \\ \sim p \\ \hline \therefore q \end{array}$$

(iii) Non-sequitur:

$$\begin{array}{l} p \\ q \\ \hline \therefore s \end{array}$$

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## Inconsistent:

A set of premises $\{P_1, P_2 \dots P_n\}$ are said to be inconsistent, if all premises cannot be simultaneously true.

i.e., $(P_1 \wedge P_2 \wedge P_3 \wedge \dots \wedge P_n) \equiv F$

In any argument, if premises are inconsistent then the argument is considered valid.

## Conditional Proof:

$((P_1 \wedge P_2 \wedge \dots P_n) \wedge Q) \underset{①}{\rightarrow} R$ is equivalent to

$(P_1 \wedge P_2 \wedge \dots \wedge P_n) \underset{②}{\rightarrow} (Q \rightarrow R)$

If the question is asked in form ② then we can convert it to the form ① and solve it.

## Indirect proof (proof by contradiction):

→ In this method we start by assuming argument is not valid.

→ So we take negation of the conclusion as a new premise.

→ when this new premise combined with other premises lead to any contradiction, then we say given argument is valid.

Eg: $a \rightarrow b$

$\sim(a \wedge b)$

---

$\therefore \sim a$

Applying indirect proof

'a' is new primise

$a \rightarrow b$

$a$

$\therefore b$

$\sim(a \wedge b)$ ~~is~~ is said to be true

But from ① & ② $a \wedge b$ is true which is a contradiction

$\therefore$ given argument is valid.

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## Functional Completeness:

A set of ~~con~~ connectives is said to be functionally complete if every ~~pos~~ propositional function can be represented used only those connectives.

Eg: $\underbrace{\{\vee, \sim\}}_{NOR}$, $\underbrace{\{\wedge, \sim\}}_{NAND}$, $\{\rightarrow, \sim\}$, are functionally complete.

It is similar to universal gates

## Normal forms:

### Disjunctive Normal form:

* It is representation of OR of ANDs

  i.e., Sum of products

### Conjunctive Normal Form:

* It is represented as AND of ORs

  i.e., Product of Sums

07/06/20

# First Order Logic (or) Predicate logic

**open statement:**

A statement for which we can't assign a truth value unless an input is provided is called open statement. when i/p is given it changes into a simple statement.

Eg: x is an even number (open statement)

2 is an even number (simple statement)

**Domain (or) ~~Discourse~~ (or) Universe of discourse (or) Domain of discourse:**

It is the set from which an i/p is given to an open statement.

→ Now consider

$$\underbrace{x}_{\substack{\text{Subject} \\ \text{(or)} \\ \text{Predicate variable}}} \; \underbrace{\text{is an even number}}_{\substack{\text{predicate} \\ P(x)}}$$

P is called predicate

x is called predicate variable.

$P(x)$: x is an even number

$P(2)$ : True $P(3)$: False

**Quantifiers:**

(i) Universal Quantifier (for all) ($\forall$):

Let $P(x)$ be a predicate

$\forall x P(x)$ is true, iff $P(x)$ is true for any x

is false, iff we can show $P(x)$ is false for atleast one x.

(ii) Existential Quantifier (there exists) ($\exists$):

~~$\exists x$ is tr~~

$\exists x P(x)$ is true, iff $P(x)$ is true for atleast one x

is false, iff $P(x)$ is false for any x.

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Note:

If the domain is empty

* $\forall x P(x)$ is true
* $\exists x P(x)$ is false

* If domain $D = \{x_1, x_2, x_3, x_4\}$

$$\forall x P(x) = P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge P(x_4)$$

$$\exists x P(x) = P(x_1) \vee P(x_2) \vee P(x_3) \vee P(x_4)$$

Negation of Quantified statements:

* $\sim[\forall x, P(x)] = \exists x, \sim P(x)$
* $\sim[\exists x, P(x)] = \forall x, \sim P(x)$

(change $\forall$ to $\exists$ and $\exists$ to $\forall$ and negate the predicate)

Equivalences & Implications:

→ $\boxed{\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)}$
$\boxed{\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)}$

→ $\boxed{(\forall x P(x)) \vee (\forall x Q(x)) \Rightarrow \forall x [P(x) \vee Q(x)]}$
$\boxed{\exists x [P(x) \wedge Q(x)] \Rightarrow \exists x P(x) \wedge \exists x Q(x)}$

for all implications here the reverse is not true

→ $\boxed{\forall x [P(x) \rightarrow Q(x)] \Rightarrow \forall x P(x) \rightarrow \forall x Q(x)}$
$\boxed{\forall x [P(x) \leftrightarrow Q(x)] \Rightarrow \forall x P(x) \leftrightarrow \forall x Q(x)}$
$\boxed{\exists x [P(x) \rightarrow Q(x)] \not\Rightarrow \forall x A(x) \rightarrow \exists x B(x)}$

Note:

* $\sim$[All true] $\equiv$ Atleast one false (or) not all true
* $\sim$[All false] $\equiv$ Atleast one true (or) not all false

# English to logical statement:

* Every lion drinks coffee

$$\forall x\, [L(x) \rightarrow C(x)]$$

* Some cats are intelligent

$$\exists x\, [cat(x) \wedge int(x)]$$

**Note:**

* all - $\forall x\, [(\ ) \rightarrow (\ )]$
* no/none - $\forall x\, [(\ ) \rightarrow \sim(\ )]$
* not all - $\exists x\, [(\ ) \wedge \sim(\ )]$

## Binding variables:

**scope:** The part of logical expression to which a quantifier is applied is called the scope of this quantifier.

Eg: $\exists x\, (P(x) \wedge Q(x)) \vee \forall x\, R(x)$

Scope of $\exists x$ is $P(x) \wedge Q(x)$

$\forall x$ is $R(x)$

**Bound:** A variable which is quantified or under scope of a variable is called bound variable.

**Free variable:** A variable which is not quantified is called free variable.

Eg: $\exists x\, (x+y=1)$

$x$ is bound variable

$y$ is free variable and doesn't change w.r.to $x$

## Restricted Equivalences:

The following equivalences hold if $x$ does not occur free in the well formed formula $C$:

* $\forall x\, C \equiv C$

  $\exists x\, C \equiv C$

* $\forall x\,(C \vee A(x)) \equiv C \vee \forall x\, A(x)$

  $\exists x\,(C \vee A(x)) \equiv C \vee \exists x\, A(x)$

* $\forall x\,(C \wedge A(x)) \equiv C \wedge \forall x\, A(x)$

  $\exists x\,(C \wedge A(x)) \equiv C \wedge \exists x\, A(x)$

* $\forall x\,(C \to A(x)) \equiv C \to \forall x\, A(x)$

  $\exists x\,(C \to A(x)) \equiv C \to \exists x\, A(x)$

  $\forall x\,(A(x) \to C) \equiv \exists x\, A(x) \to C$

  $\exists x\,(A(x) \to C) \equiv \forall x\, A(x) \to C$

Proof:

$\forall x\,(A(x) \to C)$

$\equiv \forall x\,(\sim A(x) \vee C)$

$\equiv \forall x \sim A(x) \vee C$

$\equiv \exists x\, A(x) \to C$

Slly for others.

Note:

* $\forall x\, P(x) \vee \forall x\, Q(x) \equiv \forall x \forall y\,(P(x) \vee Q(y))$

  $\forall x\, P(x) \wedge \forall x\, Q(x) \equiv \forall x \forall y\,(P(x) \wedge Q(y))$

* $\forall x\, P(x) \vee \exists x\, Q(x) \equiv \forall x \exists y\,(P(x) \vee Q(y)) \equiv \exists y \forall x\,(P(x) \vee Q(y))$

  $\forall x\, P(x) \wedge \exists x\, Q(x) \equiv \forall x \exists y\,(P(x) \wedge Q(y)) \equiv \exists y \forall x\,(P(x) \wedge Q(y))$

* $\exists x\, P(x) \vee \exists x\, Q(x) \equiv \exists x \exists y\,(P(x) \vee Q(y))$

  $\exists x\, P(x) \wedge \exists x\, Q(x) \equiv \exists x \exists y\,(P(x) \wedge Q(y))$

No need to remember. But try to prove

* $\forall x\, P(x) \to \forall x\, Q(x) \equiv \exists x \forall y\,(P(x) \to Q(y))$

  $\exists x\, P(x) \to \forall x\, Q(x) \equiv \forall x \forall y\,(P(x) \to Q(y))$

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$\forall x P(x) \rightarrow \exists x Q(x) \equiv \exists x \exists y (P(x) \rightarrow Q(y))$

$\exists x P(x) \rightarrow \exists x Q(y) \equiv \forall x \exists y (P(x) \rightarrow Q(y))$

Note:

* $\forall x A(x) \vee \forall x B(x) \equiv \forall x A(x) \vee \forall y B(y)$

As long as scopes don't overlap, variable names can be change.

## Nested Quantifiers:

(i) $\forall x \forall y \equiv \forall y \forall x$:

$\forall x \forall y P(x,y)$ is true when $P(a,b) = \text{true}$ $\forall a \in D, \forall b \in D$

is false when $P(a,b) = \text{false}$ for atleast one $(a,b) \in D \times D$

D is domain

(ii) $\exists x \exists y \equiv \exists y \exists x$

$\exists x \exists y P(x,y)$ is true when $P(a,b) = \text{true}$ for atleast one $(a,b) \in D \times D$

is false when $P(a,b) = \text{false}$ for $\forall a \in D, \forall b \in D$

(iii) $\forall x \exists y$:

$\forall x \exists y P(x,y)$ is true when

$\forall a \in A$, there exist atleast one 'b' such that $P(a,b) = T$

is false when

there, exists $a \in A$ for which no 'b' exists such that $P(a,b) = \text{true}$

~~A this kind of a function (Every x has an image~~).

* It is a kind of relation in which every $x$ is related to some $y$.

~~(iv) $\exists x \forall y$:~~

~~$\exists x \forall y\, P(x,y)$ is true when~~

(iv) $\exists y \forall x$:

$\exists y \forall x\, P(x,y)$ is true when,

you have atleast one $b \in B$ ($b$ is fixed) such that

$\forall a \in A \quad P(a,b) \equiv \text{True}$

it is false when

you can't show such fixed $y$.

* It is a kind of relation in which,

for some $y$, every $x$ relates to $y$.

Note:

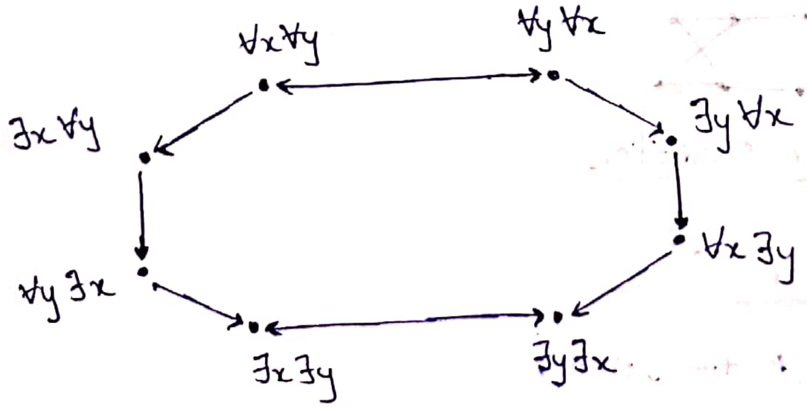
* $\forall x \forall y \equiv \forall y \forall x$

  $\exists x \exists y \equiv \exists y \exists x$

* $\exists y \forall x \Rightarrow \forall x \exists y$

  $\exists x \forall y \Rightarrow \forall y \exists x$

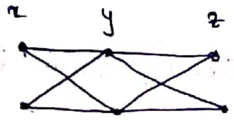
Relation b/w different nested quantifiers:



**Note:**

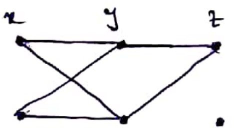
* If we have more than two nested quantifiers, we can represent as graphs and see them as if they were loops.

Eg: $\forall x \forall y \forall z$ Domain, $D = \{1, 2\}$

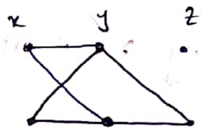


| $\forall x \forall y$ | $\forall x \forall y \forall z$ | |
|---|---|---|
| $1 < {}^{1}_{2}$ | $1-1 < {}^{1}_{2}$ | 111 |
| $2 < {}^{1}_{2}$ | $1-2 < {}^{1}_{2}$ | 112 |
| 1 − 1 | $2-1 < {}^{1}_{2}$ | 121 |
| 1 − 2 | $2-2 < {}^{1}_{2}$ | 122 |
| 2 − 1 | | 211 |
| 2 − 2 | | 212 |
| | | 221 |
| | | 222 |

Eg: $\forall x \forall y \exists z$



(or)



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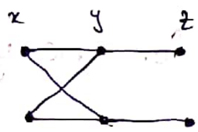
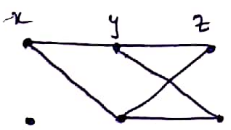


Eg: $\exists x \forall y \forall z$



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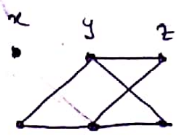
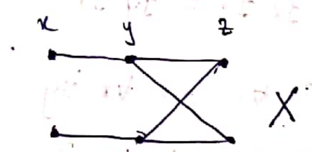
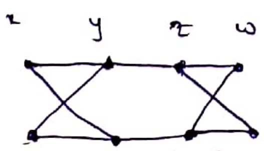




(This is not valid)

Eg: $\forall x \forall y \exists z \forall w$



## Inference Rules with Quantifiers

i) **Universal Specification (U.S)**

if $\forall x P(x)$ is true, then

$P(a)$ is true, $a \in$ Domain

a is any random element

(ii) <u>Universal Generalization</u> (U.G)

If $P(a)$ is true for some random element $a \in$ Domain

and in that way if $P(a)$ is true $\forall a \in$ Domain

then $\forall x\, P(x)$ is true

(iii) <u>Existential Specification</u> (E.S)

if $\exists x\, P(x)$ is true, then

$P(a)$ is true, $a \in$ Domain

a is a specific or fixed element.

(iv) <u>Existential Generalization</u> (E.G)

If $P(a)$ is true for some fixed $a \in A$,

then $\exists x\, P(x)$ is true.

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